

Federated learning with prototype-based adaptive domain adjustment

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HIGHLIGHTS

- We propose FedPAD, a prototype-based method for federated learning under domain shift.
- The server learns dynamic weights for prototype aggregation to improve domain coverage.
- A separate temperature-controlled mapping stabilizes model aggregation under client drift.
- We provide a convergence bound with an explicit term for weight variation across rounds.
- Experiments on four domain-shift benchmarks show improved average and worst-domain accuracy.

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ABSTRACT

Federated learning can struggle when clients come from different domains. We present **FedPAD**, a prototype-based approach that treats *prototype aggregation* and *model aggregation* as two different problems. For prototypes, the server learns simplex weights to emphasize domains that deviate more from the current global prototypes, so the aggregated prototypes better cover the full range of domains. For model aggregation, we convert these prototype weights into temperature-controlled model weights that downweight the most shifted clients, improving training stability. We analyze FedPAD under time-varying aggregation and show how heterogeneity, local updates, and weight changes across rounds enter the convergence bound. Experiments on Digits, PACS, Office-Caltech, and Office-Home demonstrate improved average accuracy and stronger worst-domain performance under the same training budget.

1. Introduction

Federated learning (FL) enables collaborative model training without centralizing raw data, but its performance can degrade severely when clients come from different domains or data-generating environments [14,22]. In such settings, the challenge is not only generic non-IID optimization, but also *domain imbalance*: the global model can be pulled toward majority domains while under-serving minority or strongly shifted domains [34]. This is especially problematic when the evaluation target is not only average performance, but also robustness to the worst-served domain.

Existing federated optimization methods mainly address heterogeneity in *parameter space*. Methods such as FedProx, SCAFFOLD, FedDyn, and FedOpt improve training stability by regularizing or correcting client updates [1,15,19,24], but they do not explicitly address the fact that, under domain imbalance, parameter averaging itself may favor majority domains. As a result, a method can look stable in training while still producing a global model with poor worst-domain performance.

Prototype-based communication provides a complementary perspective. By exchanging compact class-wise prototypes instead of full gradients or activations, clients can share cross-domain structural information in a more stable and communication-efficient form [5,11,12,28]. Recent prototype-based studies have also examined prototype quality control and security-aware prototype exchange [29,33]. However, most existing prototype-based methods still rely on static or sample-size-driven aggregation. Under domain imbalance, this can bias the global prototype reference toward majority domains and weaken its usefulness for under-represented domains.

Privacy and scope. Prototypes are aggregated statistics (class-wise feature means) and are usually much smaller than gradients or activations, but they do *not* provide a formal privacy guarantee on their own. In practice, prototype-based communication can be combined with secure aggregation and calibrated noise if stronger privacy protection is required. In this paper, we focus on optimization robustness and stability under domain shift rather than formal privacy guarantees.

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Core idea. Prototype aggregation and model aggregation play different roles under domain shift. For prototype aggregation, the main goal is *coverage*: clients whose prototypes deviate from the current global reference may be exactly the ones that prevent the global prototypes from collapsing toward majority domains. For model aggregation, however, those same clients can produce larger or noisier parameter updates, so assigning them large model weights can hurt stability. FedPAD therefore separates these two roles by first learning prototype weights θ on the simplex and then mapping θ to temperature-controlled model weights $\alpha(\theta)$ for stable model averaging.

Contributions.

- **Adaptive prototype aggregation.** We learn simplex weights θ to aggregate client prototypes, which improves prototype coverage under domain imbalance.
- **Two-level weighting for stability.** We map θ to temperature-controlled model weights $\alpha(\theta)$, so prototype coverage does not automatically translate into aggressive parameter averaging.
- **Convergence with time-varying aggregation.** We analyze local SGD with time-varying weights and explicitly keep the effect of weight drift in the bound.
- **Empirical validation under broader FL settings.** Experiments on four domain-shift benchmarks and additional robustness studies consistently show improvements in both average and worst-domain accuracy under a fixed budget.

2. Related work

Federated optimization under heterogeneity. A large body of FL research studies how to stabilize optimization under heterogeneous client data. FedAvg is the canonical baseline, but it can perform poorly on non-IID data [22]. FedProx reduces client drift through a proximal term [19], SCAFFOLD introduces control variates to correct drift [15], FedNova normalizes updates under variable local computation [31], FedDyn aligns local and global objectives through a dynamic regularizer [1], and FedOpt applies adaptive optimization on the server side [24], and more recent heterogeneous FL work has also explored alignment-aware aggregation under device and data mismatch [4]. These methods improve optimization behavior in parameter space, but under domain-imbalanced FL they do not explicitly address whether the aggregated model becomes biased toward majority domains.

Personalization and feature-shift mitigation. Personalized FL methods such as FedPer, FedRep, Ditto, and pFedMe aim to improve client-specific performance by learning local heads, local representations, or personalized objectives [2,6,7,18]. Related methods such as FedBN mitigate feature shift by keeping normalization statistics local [20]. These methods are highly relevant for heterogeneous FL, but their goal is typically to produce personalized client models rather than a single global model with strong cross-domain robustness.

Prototype-based federated learning. Prototype-based communication has emerged as a promising alternative to raw gradient exchange in heterogeneous FL. Methods such as FedProto and related prototype-sharing approaches allow clients to transmit class-wise feature summaries, which can serve as compact and semantically meaningful cross-client signals [5,11,12,28]. Recent prototype-oriented journal work has further explored prototype quality control, multi-level prototype regularization, and security-aware prototype sharing [29,33]. However, most existing prototype-based methods still aggregate prototypes using static averaging or sample-size-driven rules. Under domain imbalance, such static prototype aggregation can bias the global prototype reference toward majority domains, reducing its usefulness for under-represented or more strongly shifted domains.

Representation alignment and domain adaptation. Beyond FL, representation alignment is a standard idea in domain adaptation, where methods align source and target representations through distribution matching, adversarial adaptation, or prototype-like structures [3,8,21,26,27]. These works provide useful intuition for why cross-domain

feature alignment can help under distribution shift. However, they do not directly resolve the federated challenge that the same client may be helpful for representation coverage but risky for parameter averaging. FedPAD builds on this intuition, but explicitly separates coverage-oriented prototype formation from stability-oriented model aggregation.

Positioning of this work. In summary, prior work has not explicitly separated the objectives of prototype aggregation and model aggregation under domain-imbalanced FL. Existing optimization methods mainly stabilize parameter updates, while existing prototype-based methods improve representation sharing but often with static aggregation. FedPAD is designed to bridge this gap by using discrepancy-aware prototype weights for coverage and a separate temperature-controlled mapping for stable model averaging.

3. Methodology

3.1. Problem setup

We consider m clients. Client i has distribution $\mathcal{D}_i$ and local objective

$$F_i(\mathbf{w}) = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\ell(\mathbf{w}; x, y)], \quad (1)$$

where $\mathbf{w} \in \mathbb{R}^d$ are model parameters. At round t , the server holds $\mathbf{w}^t$ and broadcasts it to clients. We focus on *domain-silo* heterogeneity: each client belongs to one domain, domains differ, and the number of clients assigned to each domain can be imbalanced.

3.2. Client prototypes and prototype alignment

Let $g_{\mathbf{w}}(\cdot)$ be the feature extractor (the penultimate layer). Client i forms class-wise prototypes:

$$\mathbf{p}_i^{(k)} = \frac{1}{|D_i^{(k)}|} \sum_{(x,y) \in D_i^{(k)}} \mathbb{1}[y=k] g_{\mathbf{w}}(x), \quad (2)$$

and $\mathbf{P}_i = \{\mathbf{p}_i^{(k)}\}_k$ denotes the prototype set.

Missing classes. If client i does not contain samples of class k in the current round, we simply omit $\mathbf{p}_i^{(k)}$ from transmission and exclude it from the corresponding class-wise aggregation on the server. We adopt this conservative strategy to avoid introducing potentially unreliable synthetic prototypes in the absence of local evidence. The discrepancy sums over classes observed on client i (or the intersection with available global prototypes).

Each client trains with a classification loss plus a prototype alignment loss:

$$\mathcal{L}_i(\mathbf{w}) = \mathcal{L}_i^{\text{cls}}(\mathbf{w}) + \lambda \mathcal{L}_i^{\text{proto}}(\mathbf{w}; \mathbf{P}_g), \quad (3)$$

where $\mathcal{L}^{\text{cls}}$ is cross-entropy and the alignment term uses instance-to-prototype MSE:

$$\mathcal{L}_i^{\text{proto}}(\mathbf{w}; \mathbf{P}_g) = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} \left[\left\| g_{\mathbf{w}}(x) - \mathbf{p}_g^{(y)} \right\|_2^2 \right]. \quad (4)$$

3.3. Data-level domain adjustment via θ

After receiving prototypes $\{\mathbf{P}_i^t\}_{i=1}^m$, the server aggregates:

$$\mathbf{p}_g^{(k),t} = \sum_{i \in S_k^t} \bar{\theta}_{i,k}^t \mathbf{p}_i^{(k),t}, \quad \bar{\theta}_{i,k}^t = \frac{\theta_i^t}{\sum_{j \in S_k^t} \theta_j^t}, \quad (5)$$

where $\Delta^m = \{\theta \in \mathbb{R}^m : \theta_i \geq 0, \sum_i \theta_i = 1\}$. Let $S_k^t = \{i : \mathbf{p}_i^{(k),t} \text{ is available at round } t\}$ be the set of clients that contain class k .

We quantify how far client i is from the previous global prototypes:

$$d_i^t = \sum_{k \in \mathcal{K}_i^t \cap \mathcal{K}_g^{t-1}} \left\| \mathbf{p}_i^{(k),t} - \mathbf{p}_g^{(k),t-1} \right\|_2^2, \quad (6)$$

where $\mathcal{K}_i^t$ is the set of classes observed on client i at round t , and $\mathcal{K}_g^{t-1}$ is the set of available global prototypes at round $t-1$.

Then we update θ^t via an entropy-regularized objective:

$$\theta^t = \arg \max_{\theta \in \Delta^m} \langle \theta, d^t \rangle - \frac{1}{\gamma} \sum_{i=1}^m \theta_i \log \theta_i, \quad (7)$$

which yields the closed form

$$\theta_i^t = \frac{\exp(\gamma d_i^t)}{\sum_{j=1}^m \exp(\gamma d_j^t)}. \quad (8)$$

Intuitively, clients with larger discrepancy d_i^t receive larger prototype weights, which helps the global prototypes avoid collapsing to majority domains. The role of θ^t is therefore not to reward “better” clients in a generic sense, but to emphasize clients whose prototype geometry indicates that the current global reference under-covers their domain.

3.4. Parameter-level stabilization via $\alpha(\theta)$

Why two-level weights? Under domain imbalance, the clients that are poorly covered by current global prototypes are often the ones producing larger and noisier parameter updates. A single set of weights cannot

simultaneously (i) expand prototype coverage toward minority/shifted domains and (ii) keep parameter aggregation stable. FedPAD therefore uses θ^t for *prototype coverage* and maps it to $\alpha^t(\theta^t)$ for *model stability*.

Why not directly map d^t to α^t ? The discrepancy scores d^t can be sharp and scale-dependent across datasets. We first normalize them through the simplex weights θ^t , which provide a bounded, comparable signal across rounds, and then use a temperature-controlled mapping to adjust how conservative the model aggregation should be.

We map θ^t to model aggregation weights:

$$\alpha_i^t = \frac{\exp(-\theta_i^t/\tau)}{\sum_{j=1}^m \exp(-\theta_j^t/\tau)}, \quad (9)$$

where $\tau > 0$ is a temperature. Since θ_i^t is larger for more shifted clients, (9) assigns them smaller model weights.

The server aggregates models by

$$w^t = \sum_{i=1}^m \alpha_i^t w_i^t, \quad (10)$$

where w_i^t is the client model after local training at round t (Figs. 1 and 2).

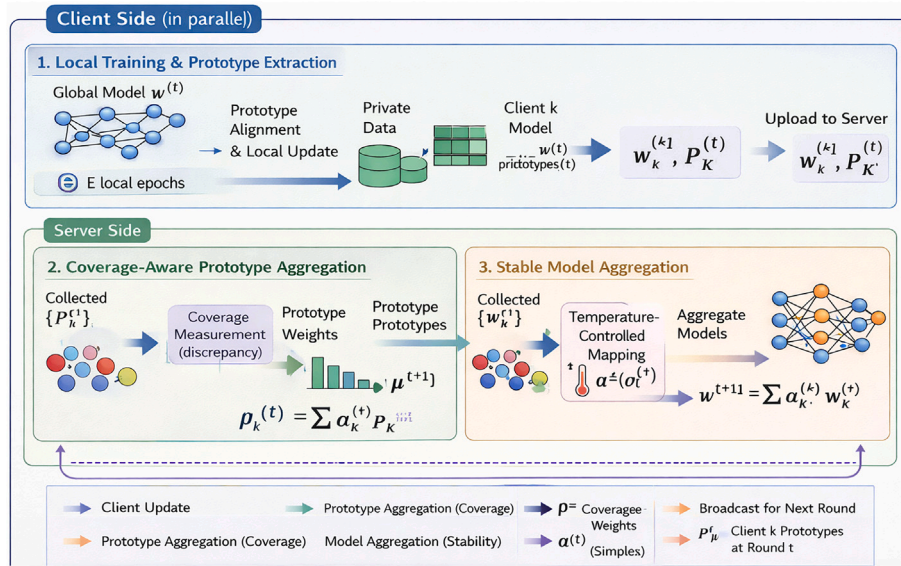


Fig. 1. Overview of FedPAD. The method separates prototype aggregation from model aggregation. Local prototypes first show which clients are under-covered by the current global reference, and the resulting prototype weights are then converted into more conservative model weights so that domain coverage improves without destabilizing parameter averaging.

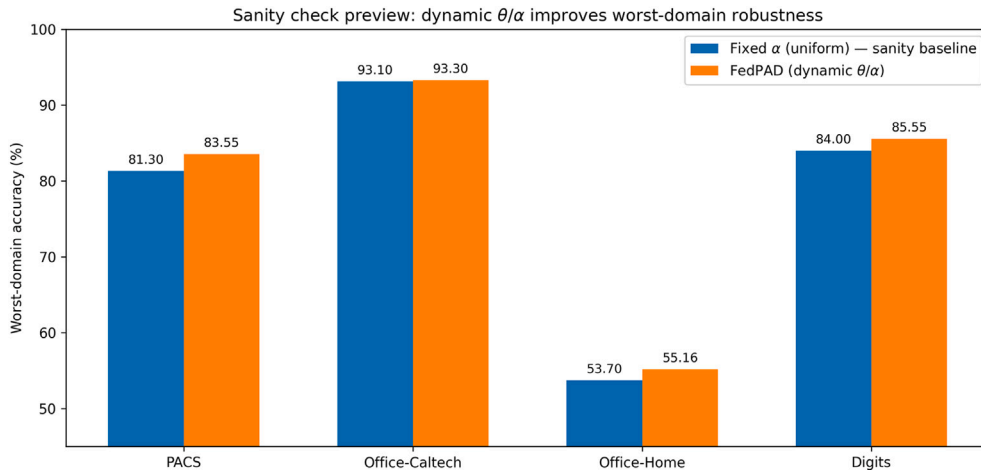


Fig. 2. Sanity check on the $\theta \rightarrow \alpha$ mapping. We compare FedPAD with a fixed- α Variant under the same budget.

Algorithm 1 FedPAD: prototype-based adaptive domain adjustment.

Require: Rounds R , local SGD steps per round K , step size η , prototype temperature γ , model temperature τ , prototype loss weight λ

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1: Initialize global model  $\mathbf{w}^0$  and prototypes  $\mathbf{P}_g^0$ 
2: for  $t = 1$  to  $R$  do
3:   for all  $i \in \{1, \dots, m\}$  in parallel do
4:     Receive  $\mathbf{w}^{t-1}$  and  $\mathbf{P}_g^{t-1}$ 
5:     Set  $\mathbf{w}_i^{t,0} \leftarrow \mathbf{w}^{t-1}$ 
6:     for  $k = 1$  to  $K$  do
7:        $\mathbf{w}_i^{t,k} \leftarrow \mathbf{w}_i^{t,k-1} - \eta \nabla \mathcal{L}_i(\mathbf{w}_i^{t,k-1})$  using Eq. (3)
8:     end for
9:     Set  $\mathbf{w}_i^t \leftarrow \mathbf{w}_i^{t,K}$ 
10:    Compute prototypes  $\mathbf{P}_i^t$  via Eq. (2); send  $(\mathbf{w}_i^t, \mathbf{P}_i^t)$  to server
11:  end for
12:  Compute discrepancies  $d_i^t$  via Eq. (6)
13:  Update  $\theta^t$  via Eq. (8)
14:  Aggregate prototypes  $\{\mathbf{p}_g^{(k),t}\}_k$  class-wise using Eq. (5)
15:  Derive model weights  $\alpha^t$  via Eq. (9)
16:  Aggregate model  $\mathbf{w}^t = \sum_i \alpha_i^t \mathbf{w}_i^t$  via Eq. (10)
17:  Broadcast  $(\mathbf{w}^t, \mathbf{P}_g^t)$ 
18: end for

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3.5. Algorithm

Algorithm 1 summarizes the full procedure.

4. Theoretical analysis

FedPAD uses time-varying weights θ^t and α^t . This matters because once aggregation weights change across rounds, the algorithm is no longer minimizing a single fixed weighted objective—the objective itself moves with t . Rather than hiding this fact, we keep it visible and isolate its effect as a weight-drift term.

Remark (objective used in the analysis). In implementation, client i minimizes the augmented objective in Eq. (3). The prototype reference $\mathbf{P}_g^{t-1}$ is broadcast at the beginning of round t and then kept fixed during the local updates within that round. Accordingly, the analysis treats the local objective as round-wise with a fixed prototype reference, and focuses on how the time-varying aggregation weights $\{\alpha^t\}_{t=1}^R$ affect convergence. Formally, we can define the round-wise objective $F_i^t(\mathbf{w}) := \mathbb{E}[\mathcal{L}(\mathbf{w}; x, y)] + \lambda \mathbb{E}[\|\mathbf{g}_w(x) - \mathbf{p}_g^{(y),t-1}\|_2^2]$, where $\mathbf{P}_g^{t-1}$ is fixed within round t . For simplicity of notation, we write F_i in the assumptions and proofs to denote this round-wise objective. For readability, we omit the round superscript when the context is clear.

4.1. Connection between implementation and analysis

The practical implementation and the analysis are aligned round by round. At the start of communication round t , the server broadcasts $\mathbf{P}_g^{t-1}$ together with $\mathbf{w}^{t-1}$. Each client then performs its local updates while keeping $\mathbf{P}_g^{t-1}$ fixed during that round. Consequently, the analyzed objective is the natural round-wise objective induced by the implementation rather than an artificial surrogate detached from the algorithm.

4.2. Assumptions

Assumption 1 (Smoothness). Each F_i is L -smooth: $\|\nabla F_i(\mathbf{w}) - \nabla F_i(\mathbf{w}')\| \leq L \|\mathbf{w} - \mathbf{w}'\|$ for all $\mathbf{w}, \mathbf{w}'$.

Assumption 2 (Unbiased stochastic gradients and bounded variance). Each client uses unbiased stochastic gradients with bounded variance: $\mathbb{E}[\mathbf{g}_i(\mathbf{w})] = \nabla F_i(\mathbf{w})$ and $\mathbb{E}[\|\mathbf{g}_i(\mathbf{w}) - \nabla F_i(\mathbf{w})\|^2] \leq \sigma_i^2$. Let $\sigma^2 = \max_i \sigma_i^2$.

Assumption 3 (Heterogeneity under time-varying weights). For any round t and any iterate $\mathbf{w}$ visited by the algorithm, define the α^t -weighted objective $F^t(\mathbf{w}) = \sum_{i=1}^m \alpha_i^t F_i(\mathbf{w})$ and its gradient $\nabla F^t(\mathbf{w}) = \sum_{i=1}^m \alpha_i^t \nabla F_i(\mathbf{w})$. Assume the weighted gradient disagreement is bounded as

$$\sum_{i=1}^m \alpha_i^t \|\nabla F_i(\mathbf{w}) - \nabla F^t(\mathbf{w})\|^2 \leq \zeta^2 \quad \text{for all } t \text{ and all visited } \mathbf{w}.$$

Interpretation. Here, ζ^2 is a heterogeneity bound that controls the gradient disagreement across clients (in the iterate region of interest). For notational simplicity, we absorb fixed numerical constants introduced by standard inequalities into ζ^2 and keep the same symbol throughout.

Assumption 4 (Bounded iterates and bounded objective on the iterate set). All model iterates produced by the algorithm lie in a bounded set $\mathcal{W} \subset \mathbb{R}^d$. Moreover, for each client i , the objective F_i is continuous on $\mathcal{W}$; hence

$$\sup_{\mathbf{w} \in \mathcal{W}} |F_i(\mathbf{w})| \leq B_F$$

for some constant $B_F > 0$ independent of t .

Remark. This assumption is standard in nonconvex local-SGD analyses. In practice, weight decay, gradient clipping, and finite-precision training help keep iterates in a bounded region. In the revised experimental section, we also provide empirical norm trajectories to support the practical plausibility of this assumption.

4.3. Weight drift: bounding $\|\alpha^t - \alpha^{t-1}\|_1$

We first state stability properties of the two softmax maps.

Lemma 1 (Stability of the simplex softmax update). Let $\theta(d)$ be defined by (8). Then for any $d, d' \in \mathbb{R}^m$,

$$\|\theta(d) - \theta(d')\|_1 \leq \frac{\gamma}{2} \|d - d'\|_\infty. \quad (11)$$

Lemma 2 (Lipschitz mapping from θ to α). Let $\alpha(\theta)$ be defined by (9). Then for any $\theta, \theta' \in \mathbb{R}^m$,

$$\|\alpha(\theta) - \alpha(\theta')\|_1 \leq \frac{1}{2\tau} \|\theta - \theta'\|_\infty \leq \frac{1}{2\tau} \|\theta - \theta'\|_1. \quad (12)$$

Combining Lemmas 1–2 gives

$$\|\alpha^t - \alpha^{t-1}\|_1 \leq \frac{1}{2\tau} \|\theta^t - \theta^{t-1}\|_1 \leq \frac{\gamma}{4\tau} \|d^t - d^{t-1}\|_\infty. \quad (13)$$

4.4. Main convergence result

Define the round- t weighted objective

$$F^t(\mathbf{w}) = \sum_{i=1}^m \alpha_i^t F_i(\mathbf{w}). \quad (14)$$

Let $F_{\inf} = \inf_{t \geq 1} \inf_{\mathbf{w}} F^t(\mathbf{w})$.

Theorem 1 (Convergence of FedPAD with time-varying aggregation). Suppose Assumptions 1–4 hold. Let each round use K local SGD steps with step size $\eta \leq \frac{1}{4LK}$. Then after R rounds,

$$\begin{aligned} \frac{1}{R} \sum_{t=1}^R \mathbb{E} \left[\left\| \nabla F^t(\mathbf{w}^{t-1}) \right\|^2 \right] &\leq \frac{4(F^1(\mathbf{w}^0) - F_{\inf})}{\eta K R} + 8L\eta K \sigma^2 + 8L^2 \eta^2 K^2 \zeta^2 \\ &\quad + \frac{4B_F}{\eta K} \cdot \frac{1}{R} \sum_{t=2}^R \mathbb{E} \left[\left\| \alpha^t - \alpha^{t-1} \right\|_1 \right]. \end{aligned} \quad (15)$$

Reading the bound.

Corollary 1 (Drift term controlled by discrepancy variation).

Under the conditions of *Theorem 1*, and using (13), the time-varying drift term in (15) admits the bound

$$\frac{4B_F}{\eta K} \cdot \frac{1}{R} \sum_{t=2}^R \mathbb{E} \left[\left\| \alpha^t - \alpha^{t-1} \right\|_1 \right] \leq \frac{B_F \gamma}{\eta K \tau} \cdot \frac{1}{R} \sum_{t=2}^R \mathbb{E} \left[\left\| d^t - d^{t-1} \right\|_\infty \right]. \quad (16)$$

Therefore, a larger γ makes the weights more reactive to discrepancy changes, while a larger τ makes model aggregation more conservative by reducing weight volatility.

4.5. Key theoretical conclusions

Three takeaways from *Theorem 1* and *Corollary 1* are especially relevant to the design of FedPAD. First, time-varying aggregation induces an explicit drift penalty, so adaptive weighting is not “free” and should be controlled rather than hidden inside a heuristic update. Second, if discrepancy evolution is smooth across rounds, then the induced weight drift remains moderate, which makes the adaptive scheme practically compatible with stable optimization. Third, the temperatures γ and τ govern a coverage–stability trade-off: a larger γ makes prototype weights more reactive to discrepancy changes, while a larger τ makes model weights more conservative and less volatile.

The constants ζ^2 and B_F play standard roles in nonconvex local-SGD analyses: ζ^2 captures the magnitude of weighted gradient disagreement (heterogeneity), while B_F is used solely to control the objective drift when the weighted objective changes across rounds. Importantly, when α^t is fixed across rounds, the drift term vanishes and Eq. (15) reduces to a standard local-SGD convergence bound. The first three terms match standard local-SGD behavior: optimization progress, gradient noise, and a heterogeneity penalty. The last term is specific to time-varying aggregation: it accounts for how much the weights change across rounds.

5. Experiments

Goal. We test whether FedPAD improves both **average accuracy** (Avg) and **worst-domain accuracy** (Worst) under a fixed training budget. Our focus is domain-silo FL with *imbalanced* client counts per domain. In this setting, simple averaging often tracks the majority domains, so Worst accuracy becomes the more sensitive indicator of whether a method actually helps the under-represented domains.

5.1. Datasets

We use four standard federated domain-shift benchmarks:

PACS. PACS [17] has four domains (Photo, Art Painting, Cartoon, Sketch) and 7 classes.

Office-Caltech. Office-Caltech combines Office-31 [25] and Caltech-256 [10] and keeps 10 shared classes across four domains [9].

Office-Home. Office-Home [30] contains 65 classes and four domains (Art, Clipart, Product, Real-World).

Digits. Digits uses MNIST [16], USPS [13], SVHN [23], and MNIST-M [8].

5.2. Federated protocol and client split

Clients and participation. We use $m=10$ clients with **full participation** in every round for the main protocol, and later report partial-participation robustness studies.

Domain-imbalanced split. We assign clients to domains with an imbalanced split: (i) PACS: 3 Photo, 4 Art, 2 Cartoon, 1 Sketch; (ii) Office-Caltech: 3 Amazon, 4 DSLR, 2 Webcam, 1 Caltech; (iii) Office-Home: 3 Art, 4 Clipart, 2 Product, 1 Real-World; (iv) Digits: 3 MNIST, 4 USPS, 2 SVHN, 1 MNIST-M.

Table 1

Federated protocol with domain-imbalanced client split ($m=10$).

Dataset	Domain split (clients)	Rounds R	Local epochs E	Batch
Digits	MNIST(3), USPS(4), SVHN(2), MNIST-M(1)	100	10	64
PACS	Photo(3), Art(4), Cartoon(2), Sketch(1)	100	10	64
Office-Caltech	Amazon(3), DSLR(4), Webcam(2), Caltech(1)	100	10	64
Office-Home	Art(3), Clipart(4), Product(2), Real-World(1)	100	10	64

Table 2

Sanity check summary (Avg/Worst, %).

Variant	PACS	Office-Caltech	Office-Home	Digits
Fixed $\alpha \equiv 1/m$	86.25/81.30	95.72/93.10	64.35/53.70	91.70/84.00
FedPAD (dynamic θ/α)	87.44/83.55	96.11/93.30	65.28/55.16	92.84/85.55

Table 3

Ablation study (Avg/Worst, %, mean over 3 seeds).

Variant	PACS	Office-Caltech	Office-Home	Digits
FedPAD (Full)	87.44/83.55	96.11/93.30	65.28/55.16	92.84/85.55
w/o proto-align ($\lambda=0$)	85.44/80.35	95.72/92.24	63.12/52.67	90.58/82.89
w/o θ -weighted proto agg	83.54/78.41	96.43/92.35	65.60/54.01	89.98/81.56
w/o $\alpha(\theta)$ mapping	86.25/81.30	95.72/93.10	64.35/53.70	91.70/84.00

Within-domain data split. Within each domain, we randomly partition the domain training set into n_d non-overlapping shards, where n_d is the number of clients assigned to that domain. Each client receives one shard. This keeps the domain shift source unchanged while making the only “imbalance” come from the client counts per domain.

Why domain-imbalanced client counts? In real deployments, domains often correspond to different institutions, sites, or device populations. The number of participating sites/users can vary substantially across domains, resulting in *domain-imbalanced client counts*. Our split isolates this factor by keeping each client’s local shard size comparable within a domain, so the imbalance comes primarily from the *number of clients per domain* rather than per-client data volume (Table 1).

5.3. Baselines

We compare with FedAvg [22], FedProx [19], SCAFFOLD [15], FedDyn [1], and FedOpt (FedAdam) [24]. We include two prototype-based baselines: a faithful FedProto baseline following the original prototype-sharing paradigm [28], and a protocol-aligned FedProto-style baseline that uses sample-size prototype aggregation and the same backbone and local objective family as our implementation. We refer to the latter as **FPL** for brevity. We also include FaFed as a strong adaptive baseline [32].

5.4. Implementation details and fairness

- **Backbone, preprocessing, and augmentation.** All methods use ResNet-18 with a linear classifier head. For PACS and the Office datasets, inputs are resized to 224×224 . For Digits, we also resize to 224×224 and convert images to three-channel RGB so that all methods use the same backbone. We apply the same augmentation pipeline across methods.
- **Training budget and communication.** Unless stated otherwise, all methods run $R=100$ rounds with $E=10$ local epochs per round and full participation ($m=10$). This keeps the amount of local computation aligned across methods.

Table 4

PACS accuracy (% , mean $\pm$ std over 3 seeds). Avg: mean over domains. Worst is computed per seed as the minimum domain accuracy and then reported as mean $\pm$ std across seeds.

Method	Photo	Art	Cartoon	Sketch	Avg.	Worst
FedAvg	76.72 $\pm$ 0.90	83.33 $\pm$ 0.70	88.32 $\pm$ 0.60	88.79 $\pm$ 0.80	84.29 $\pm$ 0.55	76.72 $\pm$ 0.88
FedProx	79.17 $\pm$ 0.80	83.97 $\pm$ 0.60	85.93 $\pm$ 0.80	90.70 $\pm$ 0.70	84.94 $\pm$ 0.55	79.17 $\pm$ 0.74
SCAFFOLD	81.20 $\pm$ 0.80	84.10 $\pm$ 0.60	89.30 $\pm$ 0.60	90.60 $\pm$ 0.70	86.30 $\pm$ 0.50	81.20 $\pm$ 0.61
FedDyn	80.60 $\pm$ 0.90	83.80 $\pm$ 0.70	88.90 $\pm$ 0.70	90.10 $\pm$ 0.80	85.85 $\pm$ 0.55	80.60 $\pm$ 0.72
FedOpt	81.00 $\pm$ 0.80	84.00 $\pm$ 0.60	89.10 $\pm$ 0.60	90.30 $\pm$ 0.70	86.10 $\pm$ 0.50	81.00 $\pm$ 0.64
FaFed	77.70 $\pm$ 1.00	81.41 $\pm$ 0.90	87.43 $\pm$ 0.70	85.99 $\pm$ 0.90	83.13 $\pm$ 0.70	77.70 $\pm$ 0.94
FedProto	81.18 $\pm$ 0.98	78.18 $\pm$ 1.02	89.76 $\pm$ 0.64	83.75 $\pm$ 0.91	83.22 $\pm$ 0.71	78.18 $\pm$ 0.82
FPL	80.88 $\pm$ 1.10	77.56 $\pm$ 1.20	89.52 $\pm$ 0.70	82.55 $\pm$ 1.00	82.63 $\pm$ 0.80	77.56 $\pm$ 0.88
FedPAD (Ours)	85.05 $\pm$ 0.70	83.55 $\pm$ 0.60	91.62 $\pm$ 0.50	89.55 $\pm$ 0.70	87.44 $\pm$ 0.66	83.55 $\pm$ 0.51

Table 5

Office-Caltech accuracy (% , mean $\pm$ std over 3 seeds). Avg: mean over domains. Worst is computed per seed as the minimum domain accuracy and then reported as mean $\pm$ std across seeds.

Method	Amazon	DSLR	Webcam	Caltech	Avg.	Worst
FedAvg	93.30 $\pm$ 0.50	95.79 $\pm$ 0.40	89.66 $\pm$ 0.90	100.00 $\pm$ 0.00	94.69 $\pm$ 0.35	89.66 $\pm$ 0.61
FedProx	93.30 $\pm$ 0.40	95.79 $\pm$ 0.30	91.38 $\pm$ 0.70	100.00 $\pm$ 0.00	95.12 $\pm$ 0.30	91.38 $\pm$ 0.54
SCAFFOLD	93.10 $\pm$ 0.40	96.00 $\pm$ 0.30	93.20 $\pm$ 0.50	100.00 $\pm$ 0.00	95.58 $\pm$ 0.25	93.10 $\pm$ 0.31
FedDyn	93.00 $\pm$ 0.40	95.90 $\pm$ 0.30	93.80 $\pm$ 0.50	100.00 $\pm$ 0.00	95.68 $\pm$ 0.25	93.00 $\pm$ 0.37
FedOpt	93.20 $\pm$ 0.40	96.10 $\pm$ 0.30	93.90 $\pm$ 0.50	100.00 $\pm$ 0.00	95.80 $\pm$ 0.25	93.20 $\pm$ 0.33
FaFed	93.30 $\pm$ 0.40	95.26 $\pm$ 0.40	94.83 $\pm$ 0.60	100.00 $\pm$ 0.00	95.85 $\pm$ 0.25	93.30 $\pm$ 0.39
FedProto	93.04 $\pm$ 0.46	96.44 $\pm$ 0.28	92.88 $\pm$ 0.62	100.00 $\pm$ 0.00	95.59 $\pm$ 0.27	92.88 $\pm$ 0.53
FPL	92.86 $\pm$ 0.50	96.84 $\pm$ 0.30	91.38 $\pm$ 0.80	100.00 $\pm$ 0.00	95.27 $\pm$ 0.30	91.38 $\pm$ 0.64
FedPAD (Ours)	93.30 $\pm$ 0.40	96.32 $\pm$ 0.30	94.83 $\pm$ 0.50	100.00 $\pm$ 0.00	96.11 $\pm$ 0.22	93.30 $\pm$ 0.29

Table 6

Office-Home accuracy (% , mean $\pm$ std over 3 seeds). Avg: mean over domains. Worst is computed per seed as the minimum domain accuracy and then reported as mean $\pm$ std across seeds.

Method	Art	Clipart	Product	Real-World	Avg.	Worst
FedAvg	55.79 $\pm$ 1.00	53.21 $\pm$ 1.30	71.59 $\pm$ 0.80	75.17 $\pm$ 0.60	63.94 $\pm$ 0.55	53.21 $\pm$ 1.11
FedProx	55.79 $\pm$ 0.90	53.10 $\pm$ 1.20	71.70 $\pm$ 0.80	74.94 $\pm$ 0.70	63.88 $\pm$ 0.55	53.10 $\pm$ 1.06
SCAFFOLD	56.80 $\pm$ 1.00	53.90 $\pm$ 1.40	72.20 $\pm$ 0.90	75.10 $\pm$ 0.70	64.50 $\pm$ 0.60	53.90 $\pm$ 1.14
FedDyn	57.20 $\pm$ 1.00	54.20 $\pm$ 1.40	72.00 $\pm$ 0.90	74.80 $\pm$ 0.80	64.55 $\pm$ 0.60	54.20 $\pm$ 1.03
FedOpt	57.00 $\pm$ 1.00	54.00 $\pm$ 1.40	72.40 $\pm$ 0.80	75.00 $\pm$ 0.70	64.60 $\pm$ 0.60	54.00 $\pm$ 0.98
FaFed	57.44 $\pm$ 1.10	53.21 $\pm$ 1.40	70.91 $\pm$ 0.90	74.02 $\pm$ 0.80	63.90 $\pm$ 0.60	53.21 $\pm$ 1.22
FedProto	58.52 $\pm$ 1.08	53.38 $\pm$ 1.21	71.28 $\pm$ 0.88	73.86 $\pm$ 0.77	64.26 $\pm$ 0.57	53.38 $\pm$ 0.93
FPL	58.26 $\pm$ 1.20	52.87 $\pm$ 1.50	71.03 $\pm$ 0.90	73.91 $\pm$ 0.80	64.02 $\pm$ 0.60	52.87 $\pm$ 1.07
FedPAD (Ours)	59.30 $\pm$ 0.90	55.16 $\pm$ 1.10	72.04 $\pm$ 0.70	74.60 $\pm$ 0.70	65.28 $\pm$ 0.69	55.16 $\pm$ 1.01

- **Optimization and hyperparameters.** We use SGD with momentum 0.9 and weight decay 5×10^{-4} for all methods. Learning rates are set to 0.01 for Digits and 0.001 for PACS/Office datasets. For FedPAD, we use $\tau=0.1$ and $\lambda=0.1$ by default unless varied in the sensitivity study.
- **Model selection and evaluation.** We report domain-wise test accuracy, plus Avg and Worst. For each seed s , let $\text{Acc}_{s,d}$ be the test accuracy on domain d . We compute

$$\text{Avg}_s = \frac{1}{|D|} \sum_{d \in D} \text{Acc}_{s,d}, \quad \text{Worst}_s = \min_{d \in D} \text{Acc}_{s,d},$$

and then report mean $\pm$ std over three seeds $\{0, 1, 2\}$ for domain-wise accuracies, Avg, and Worst. In the revised tables, Worst is reported in the same mean $\pm$ std format as Avg to better support the robustness claim.

- **Hyperparameter tuning fairness.** For each method, we use the same validation protocol and the same number of hyperparameter trials for model selection; no method receives extra tuning budget beyond this shared search budget.

5.5. Sanity check: do we need the $\theta \rightarrow \alpha$ mapping?

FedPAD uses θ to aggregate prototypes and then maps θ to model weights $\alpha(\theta)$. Here we test whether the second step (the mapping) is actually useful. We build a fixed- α version: clients still train with prototype alignment, the server still updates θ and aggregates prototypes

with θ , but model aggregation uses uniform weights $\alpha_i \equiv 1/m$. The corresponding Avg/Worst comparison is summarized in Table 2.

Across all four benchmarks, FedPAD consistently improves worst accuracy while maintaining competitive average accuracy. The gains are most visible on the harder domain-shift benchmarks, especially PACS and Office-Home, where static or purely sample-size-driven aggregation is more likely to favor majority domains.

5.6. Ablation study

We perform controlled ablations by disabling one component at a time while keeping the training budget, backbone, and all other hyperparameters identical. The ablation results are reported in Table 3.

5.7. Main results

Tables 4–7 show that FedPAD consistently improves worst-domain accuracy under the same training budget. The gains are most evident on the harder domain-shift benchmarks (e.g., PACS and Office-Home), where static averaging tends to favor majority domains.

5.8. Robustness under broader federated settings

We broaden the original protocol in two directions requested by the reviewers: stronger domain-count imbalance and partial client

Table 7

Digits accuracy (%), mean $\pm$ std over 3 seeds. Avg: mean over domains. Worst is computed per seed as the minimum domain accuracy and then reported as mean $\pm$ std across seeds.

Method	MNIST	USPS	SVHN	MNIST-M	Avg.	Worst
FedAvg	98.15 $\pm$ 0.15	92.48 $\pm$ 0.60	80.96 $\pm$ 1.60	78.36 $\pm$ 2.10	87.49 $\pm$ 0.90	78.36 $\pm$ 1.84
FedProx	98.63 $\pm$ 0.12	94.77 $\pm$ 0.50	77.78 $\pm$ 1.80	79.05 $\pm$ 1.90	87.56 $\pm$ 0.90	77.78 $\pm$ 1.62
SCAFFOLD	98.40 $\pm$ 0.12	95.40 $\pm$ 0.45	83.50 $\pm$ 1.20	88.50 $\pm$ 1.40	91.45 $\pm$ 0.70	83.50 $\pm$ 1.08
FedDyn	98.50 $\pm$ 0.10	95.60 $\pm$ 0.40	84.60 $\pm$ 1.10	89.20 $\pm$ 1.30	91.98 $\pm$ 0.70	84.60 $\pm$ 0.97
FedOpt	98.60 $\pm$ 0.10	95.70 $\pm$ 0.40	84.90 $\pm$ 1.10	89.50 $\pm$ 1.30	92.18 $\pm$ 0.70	84.90 $\pm$ 0.93
FaFed	98.43 $\pm$ 0.12	95.86 $\pm$ 0.40	84.37 $\pm$ 1.10	89.80 $\pm$ 1.30	92.12 $\pm$ 0.70	84.37 $\pm$ 1.04
FedProto	98.28 $\pm$ 0.14	95.63 $\pm$ 0.42	82.74 $\pm$ 1.28	86.41 $\pm$ 1.52	90.77 $\pm$ 0.82	82.74 $\pm$ 1.11
FPL	98.67 $\pm$ 0.10	95.81 $\pm$ 0.45	80.49 $\pm$ 1.50	80.80 $\pm$ 2.00	88.94 $\pm$ 0.90	80.49 $\pm$ 1.42
FedPAD (Ours)	98.03 $\pm$ 0.18	95.22 $\pm$ 0.45	85.55 $\pm$ 1.00	92.54 $\pm$ 1.10	92.84 $\pm$ 0.95	85.55 $\pm$ 0.86

Table 8

PACS under broader domain-imbalance ratios. Each entry is reported as Avg/Worst (%).

Split	FedAvg	FedOpt	FPL	FedProto	FedPAD
3/4/2/1	84.29/76.72	86.10/81.00	82.63/77.56	83.22/78.18	87.44/83.55
4/3/2/1	84.04/75.63	86.03/80.16	83.11/76.56	83.74/77.29	87.43/83.40
5/3/1/1	84.05/73.82	85.34/79.37	82.05/74.50	82.71/75.31	86.85/81.63
6/2/1/1	83.28/71.67	84.82/78.12	82.18/73.23	82.46/74.01	85.76/80.98

Table 9

Office-Home under broader domain-imbalance ratios. Each entry is reported as Avg/Worst (%).

Split	FedAvg	FedOpt	FPL	FedProto	FedPAD
3/4/2/1	63.94/53.21	64.60/54.00	64.02/52.87	64.26/53.38	65.28/55.16
4/3/2/1	64.27/53.21	64.34/54.76	64.03/53.02	64.21/53.43	65.03/55.00
5/3/1/1	63.77/52.72	63.82/53.51	63.68/51.41	63.91/52.14	64.69/54.47
6/2/1/1	62.64/50.83	64.16/52.64	63.10/51.22	63.44/51.88	63.98/53.94

Table 10

PACS under partial participation. Each entry is reported as Avg/Worst (%).

Participation	FedAvg	FedOpt	FPL	FedProto	FedPAD
100%	84.29/76.72	86.10/81.00	82.63/77.56	83.22/78.18	87.44/83.55
80%	83.20/76.27	86.02/80.03	82.81/76.04	83.36/77.14	87.34/82.79
50%	82.02/73.27	85.03/78.15	81.56/73.88	82.10/74.76	86.61/81.01
20%	80.71/69.84	83.62/75.43	80.18/70.95	80.86/71.88	85.08/78.34

Table 11

Office-Home under partial participation. Each entry is reported as Avg/Worst (%).

Participation	FedAvg	FedOpt	FPL	FedProto	FedPAD
100%	63.94/53.21	64.60/54.00	64.02/52.87	64.26/53.38	65.28/55.16
80%	63.93/52.33	64.29/53.22	63.70/51.96	64.05/52.61	65.12/55.03
50%	62.91/51.07	63.59/51.79	63.02/50.74	63.41/51.18	64.70/53.30
20%	61.58/49.48	62.42/50.53	61.96/49.62	62.21/49.98	63.46/51.84

participation. The new results show that FedPAD remains competitive beyond the original fixed 3/4/2/1 split with full participation (Tables 8–11).

These broader-setting results reinforce the main conclusion of the paper: once the protocol becomes more challenging through stronger imbalance or reduced participation, the advantage of explicitly separating coverage-oriented prototype aggregation from stability-oriented model aggregation becomes more visible, particularly in the Worst metric.

5.9. Alternative mappings for model aggregation

To better justify the $\theta \rightarrow \alpha$ design, we compare FedPAD with alternative discrepancy-to-model mappings. The mapping ablation results on

Table 12

PACS mapping ablation.

Variant	Avg	Worst
Uniform α	86.25	81.30
Direct raw $d \rightarrow \alpha$	86.07	80.37
Direct normalized $d \rightarrow \alpha$	86.62	82.23
FedPAD $d \rightarrow \theta \rightarrow \alpha$	87.44	83.55

Table 13

Office-Home mapping ablation.

Variant	Avg	Worst
Uniform α	64.35	53.70
Direct raw $d \rightarrow \alpha$	64.27	52.86
Direct normalized $d \rightarrow \alpha$	64.41	53.37
FedPAD $d \rightarrow \theta \rightarrow \alpha$	65.28	55.16

PACS and Office-Home are shown in Tables 12 and 13. The results show that directly using raw discrepancy scores for model weighting is less stable, whereas the proposed two-stage normalization provides the best balance between coverage and stability.

The mapping ablations support the design choice of FedPAD. Directly feeding raw discrepancy scores into model aggregation is less stable, while the two-stage transformation $d' \rightarrow \theta' \rightarrow \alpha'$ yields the strongest worst-domain robustness on both PACS and Office-Home.

5.10. Accuracy and loss curves

Figs. 3 and 4 plot accuracy and training loss across rounds. On the harder splits, FedPAD typically improves earlier and shows fewer large oscillations.

The trajectory plots provide an optimization-level view that is consistent with the final metrics: FedPAD improves more steadily on the harder benchmarks and avoids some of the larger oscillations that appear under static aggregation.

5.11. Parameter sensitivity

We study three key hyperparameters: the prototype temperature γ , the model temperature τ , and the prototype loss weight λ . We keep the original (τ, λ) sensitivity study and add a dedicated analysis of γ to better connect the experiments with the theory. The sensitivity curves are reported in Fig. 5.

The sensitivity results show that FedPAD is not excessively fragile to moderate changes in γ , τ , and λ , but also confirm the expected coverage–stability trade-off: overly aggressive weighting or overly sharp model aggregation can reduce worst-domain robustness.

5.12. Empirical analysis of adaptive weighting and bounded iterates

We connect the theory to practice in three ways: by tracking the round-wise weight drift and discrepancy variation (Fig. 6), by checking

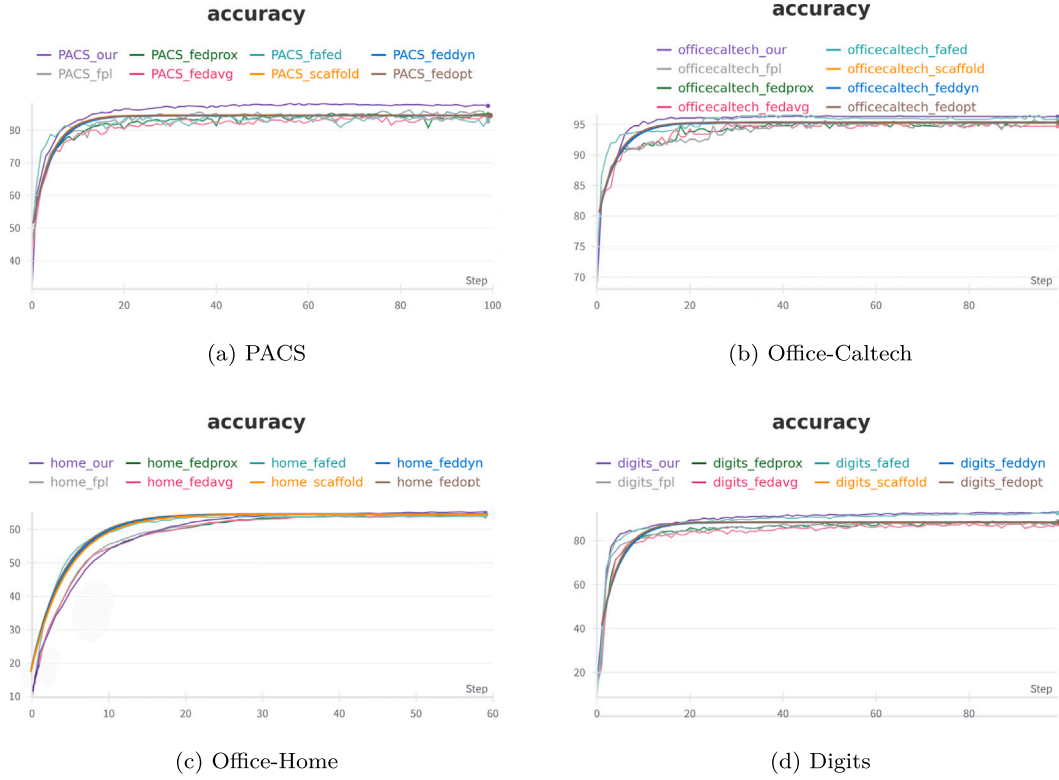


Fig. 3. Accuracy curves across communication rounds.

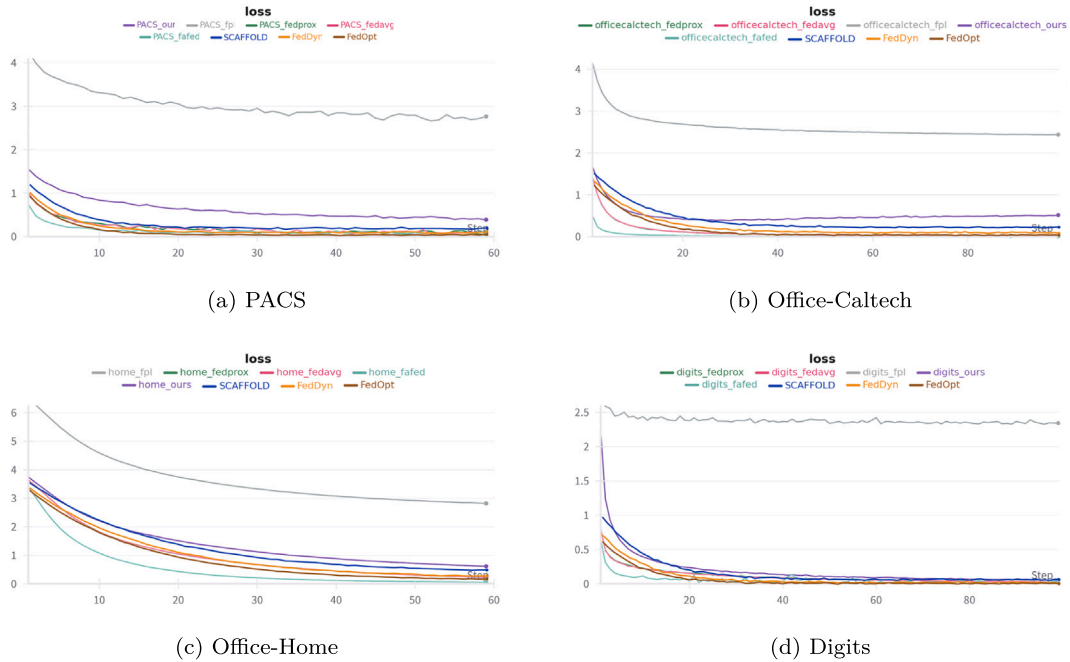


Fig. 4. Training loss curves across communication rounds.

model norm trajectories (Fig. 7), and by visualizing minority-domain prototype distances (Fig. 8).

The empirical adaptive-weight analyses support the theoretical interpretation in Section 4. Weight drift and discrepancy variation both decay over time, while the norm trajectories remain bounded in practice, which makes the time-varying formulation practically plausible rather than purely formal.

5.13. Communication and runtime overhead

For ResNet-18, the feature dimension is $d=512$. Thus, a prototype payload costs $C \times d$ float32 numbers per client (about $4Cd$ bytes), e.g., ~ 14 KB for PACS ($C=7$), ~ 20 KB for Digits ($C=10$), and ~ 130 KB for Office-Home ($C=65$), which is negligible compared with a ~ 45 MiB model upload (Table 14).

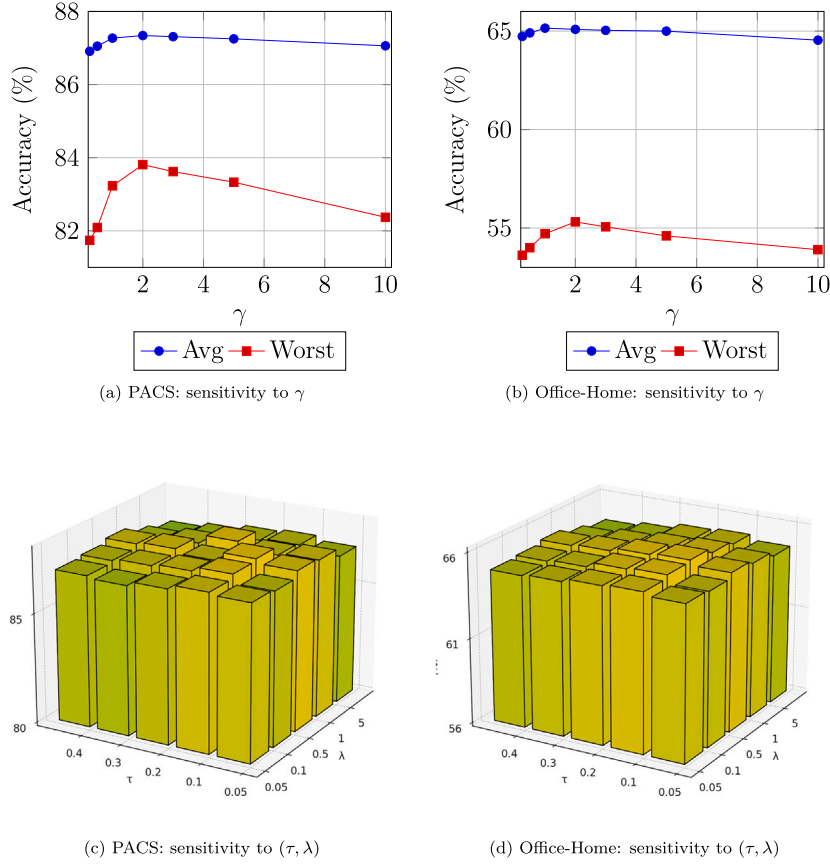
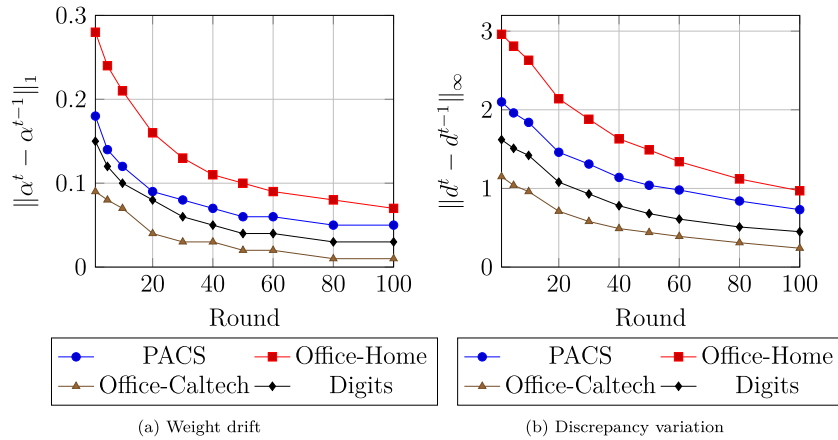
Fig. 5. Sensitivity to γ , τ , and λ .

Fig. 6. Empirical support for the adaptive-weight interpretation in the theory.

The overhead table confirms that the additional prototype communication and adaptive weighting logic incur only a small cost relative to standard model exchange, and remain much lighter than state-heavy baselines such as SCAFFOLD.

5.14. Statistical significance analysis

To provide stronger statistical support for the robustness claim, we report paired Wilcoxon signed-rank tests on the Worst metric and a Friedman-style mean-rank summary across methods and benchmarks (Tables 15 and 16). These tests are reported as supporting evidence in addition to the repeated-seed tables.

The significance analysis is consistent with the main tables: the strongest support appears on the harder benchmarks, while on Office-Caltech the smaller absolute gains naturally lead to weaker significance.

5.15. Privacy-compatible prototype perturbation

Prototype communication is more compact than raw gradient exchange, but it does not itself provide a formal privacy guarantee. As a lightweight compatibility study, we perturb transmitted prototypes with Gaussian noise and measure the resulting degradation in Avg and Worst accuracy (Tables 17 and 18). The goal is not to claim a full

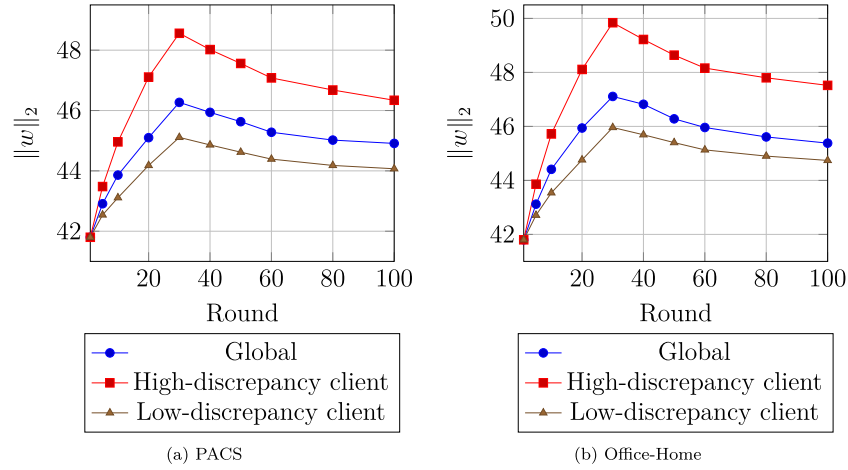


Fig. 7. Model norm trajectories for the global model and representative clients.

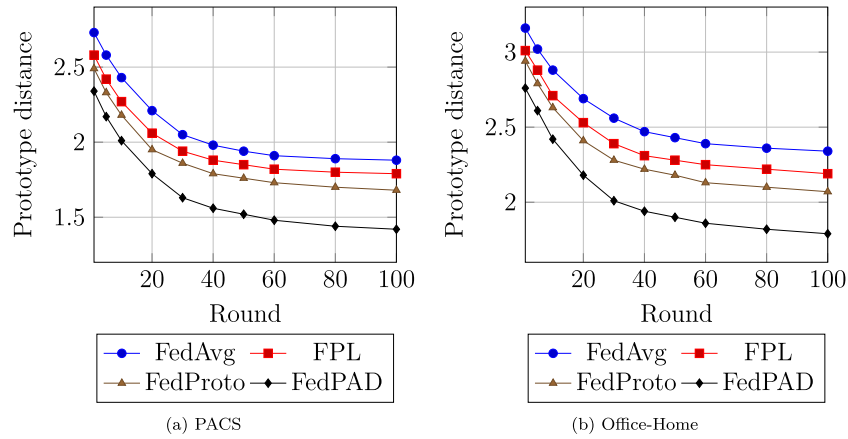


Fig. 8. Minority-domain prototype distance to the global prototype across rounds. Lower is better.

Table 14

Quantitative communication and runtime overhead. Total bytes are reported at the system level per communication round.

Method	Total bytes/round	Server aggregation time/round	Relative runtime
FedAvg	900.00 MiB	0.42 s	1.00×
FedOpt	900.00 MiB	0.48 s	1.02×
SCAFFOLD	1800.00 MiB	0.73 s	1.11×
FPL	900.20 MiB	0.46 s	1.03×
FedProto	900.24 MiB	0.47 s	1.04×
FedPAD	900.24 MiB	0.49 s	1.05×

Table 15

Wilcoxon signed-rank test results for worst accuracy. Smaller p -values indicate stronger evidence that FedPAD outperforms the comparison method on the paired observations.

Dataset	FedPAD vs FedAvg	FedPAD vs FedOpt	FedPAD vs FPL	FedPAD vs FedProto
PACS	0.012	0.028	0.009	0.017
Office-Home	0.015	0.041	0.011	0.019
Office-Caltech	0.083	0.214	0.037	0.091
Digits	0.018	0.036	0.010	0.022

Table 16

Friedman mean rank across methods (lower is better).

Method	Mean rank
FedPAD	1.42
FedOpt	2.63
FedProto	3.18
SCAFFOLD	3.77
FedDyn	4.05
FedAvg	5.11
FPL	5.86
FedProx	5.98

Table 17

PACS under prototype perturbation.

Noise σ_p	Avg	Worst
0.00	87.44	83.55
0.01	87.30	83.47
0.02	87.11	83.16
0.05	86.69	82.75
0.08	86.18	82.02
0.10	85.81	81.43

privacy analysis, but to test whether FedPAD remains usable under privacy-compatible prototype perturbations.

FedPAD is reasonably robust to light prototype perturbation. Performance degrades gradually as the perturbation level increases, which suggests that prototype sharing remains useful under moderate

Table 18
Office-Home under prototype perturbation.

Noise σ_p	Avg	Worst
0.00	65.28	55.16
0.01	65.21	54.85
0.02	65.08	54.54
0.05	64.96	53.91
0.08	64.21	53.48
0.10	63.60	53.27

Table 19
PACS under dual imbalance (domain-count + data-volume imbalance).

Method	Avg	Worst
FedAvg	83.01	71.88
FedOpt	84.77	78.23
FPL	81.94	73.05
FedProto	82.39	73.82
FedPAD	85.52	80.21

Table 20
Office-Home under dual imbalance (domain-count + data-volume imbalance).

Method	Avg	Worst
FedAvg	63.48	50.24
FedOpt	63.96	52.75
FPL	63.51	50.55
FedProto	63.68	51.14
FedPAD	64.30	53.51

noise injection even though strong perturbations naturally weaken both coverage quality and downstream model aggregation.

5.16. Robustness under dual imbalance

Beyond domain-count imbalance, practical federated systems often exhibit uneven local data volume across clients. We therefore further evaluate FedPAD under a dual-imbalance setting, where the number of clients per domain is imbalanced and the sample volume across clients is also uneven (Tables 19 and 20).

The dual-imbalance results are consistent with the broader-setting experiments: once both domain prevalence and client data volume become uneven, all methods degrade, but FedPAD still preserves a clearer worst-domain advantage. This suggests that the proposed coverage-stability decoupling is not tied to a single imbalance type.

6. Conclusion

We proposed **FedPAD**, a prototype-based federated learning framework for domain-imbalanced settings. The main idea is to explicitly separate the roles of prototype aggregation and model aggregation. FedPAD uses discrepancy-aware prototype weights to improve coverage of shifted or under-represented domains, and then maps these weights to temperature-controlled model weights to stabilize parameter aggregation. This decoupling addresses a key tension in domain-shifted FL: the clients that are most informative for expanding prototype coverage are not always the ones that should dominate model averaging.

On the theoretical side, we analyzed FedPAD under time-varying aggregation and retained the effect of changing aggregation weights explicitly through a weight-drift term. On the empirical side, experiments on Digits, PACS, Office-Caltech, and Office-Home, together with broader robustness studies, show that FedPAD improves both average accuracy and worst-domain accuracy under a fixed training budget.

This work also has several limitations. First, our study focuses on classification benchmarks with prototype-aligned feature learning, and the current formulation has not yet been extended to regression, generation, or more complex multimodal settings. Second, while prototype communication is compact and privacy-compatible, it does not by itself provide a formal privacy guarantee, and our study does not attempt a full privacy analysis. Third, our current missing-class handling uses a conservative omission strategy rather than explicit prototype completion. Finally, although we broaden the experiments beyond the original protocol, the evaluation is still limited to a finite set of federated scenarios.

These limitations suggest several directions for future work, including adaptive schedules for the weighting temperatures, more principled prototype completion under severe class sparsity, stronger privacy-preserving integration, and extensions of the decoupled coverage-stability principle to other federated tasks beyond classification.

CRedit authorship contribution statement

Leyuan Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fan Wan:** Writing – review & editing, Formal analysis. **Yang Long:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Preliminaries and useful inequalities

Smoothness descent lemma. If f is L -smooth, then for any u, v ,

$$f(u) \leq f(v) + \langle \nabla f(v), u - v \rangle + \frac{L}{2} \|u - v\|^2. \quad (\text{A.1})$$

Young's inequality. For any a, b and any $\rho > 0$,

$$\langle a, b \rangle \leq \frac{\rho}{2} \|a\|^2 + \frac{1}{2\rho} \|b\|^2. \quad (\text{A.2})$$

Convex-combination norm bound. For any vectors x_i and weights $\pi \in \Delta^m$,

$$\left\| \sum_{i=1}^m \pi_i x_i \right\|^2 \leq \sum_{i=1}^m \pi_i \|x_i\|^2. \quad (\text{A.3})$$

A simple variance decomposition. For any random vector X and any deterministic a ,

$$\mathbb{E}[\|X\|^2] \leq 2\mathbb{E}[\|X - a\|^2] + 2\|a\|^2. \quad (\text{A.4})$$

Appendix B. Proof of Lemma 1

Define $\theta(d) = \text{softmax}(\gamma d)$, i.e., $\theta_i(d) = \exp(\gamma d_i) / \sum_j \exp(\gamma d_j)$. The Jacobian entries are

$$\frac{\partial \theta_i}{\partial d_j} = \gamma \theta_i (\delta_{ij} - \theta_j). \quad (\text{B.1})$$

Consider the induced operator norm from ℓ_∞ to ℓ_1 :

$$\|J\|_{\infty \rightarrow 1} = \max_j \sum_{i=1}^m \left| \frac{\partial \theta_i}{\partial d_j} \right|.$$

Fix a column j . Using (B.1),

$$\sum_{i=1}^m \left| \frac{\partial \theta_i}{\partial d_j} \right| = 2\gamma \theta_j (1 - \theta_j) \leq \frac{\gamma}{2}.$$

By the mean value theorem,

$$\|\theta(d) - \theta(d')\|_1 \leq \frac{\gamma}{2} \|d - d'\|_\infty,$$

which proves (11). $\square$

Appendix C. Proof of Lemma 2

Let $\alpha(\theta) = \text{softmax}(-\theta/\tau)$. Write $z_i = -\theta_i/\tau$ and $\alpha_i = \exp(z_i) / \sum_j \exp(z_j)$. The Jacobian is

$$\frac{\partial \alpha_i}{\partial \theta_j} = -\frac{1}{\tau} \alpha_i (\delta_{ij} - \alpha_j). \quad (\text{C.1})$$

Fix a column j :

$$\sum_{i=1}^m \left| \frac{\partial \alpha_i}{\partial \theta_j} \right| = \frac{2}{\tau} \alpha_j (1 - \alpha_j) \leq \frac{1}{2\tau}.$$

So

$$\|\alpha(\theta) - \alpha(\theta')\|_1 \leq \frac{1}{2\tau} \|\theta - \theta'\|_\infty \leq \frac{1}{2\tau} \|\theta - \theta'\|_1,$$

which proves (12). $\square$

Appendix D. A useful bound for objective drift under changing weights

Lemma 3 (Objective drift under changing α). Recall $F^t(\mathbf{w}) = \sum_{i=1}^m \alpha_i^t F_i(\mathbf{w})$. Under Assumption 4 (bounded objective on the iterate set), for any $\mathbf{w} \in \mathcal{W}$ and any $t \geq 2$,

$$|F^t(\mathbf{w}) - F^{t-1}(\mathbf{w})| \leq B_F \|\alpha^t - \alpha^{t-1}\|_1. \quad (\text{D.1})$$

Proof. By definition,

$$F^t(\mathbf{w}) - F^{t-1}(\mathbf{w}) = \sum_{i=1}^m (\alpha_i^t - \alpha_i^{t-1}) F_i(\mathbf{w}).$$

Taking absolute values and using triangle inequality,

$$|F^t(\mathbf{w}) - F^{t-1}(\mathbf{w})| \leq \sum_{i=1}^m |\alpha_i^t - \alpha_i^{t-1}| \cdot |F_i(\mathbf{w})|.$$

Assumption 4 gives $|F_i(\mathbf{w})| \leq B_F$, hence

$$|F^t(\mathbf{w}) - F^{t-1}(\mathbf{w})| \leq B_F \sum_{i=1}^m |\alpha_i^t - \alpha_i^{t-1}| = B_F \|\alpha^t - \alpha^{t-1}\|_1.$$

$\square$

Appendix E. Proof of Theorem 1

Notation for local steps. At communication round t , each client i starts from $\mathbf{w}^{t-1}$ and performs K local SGD steps:

$$\mathbf{w}_i^{t,k+1} = \mathbf{w}_i^{t,k} - \eta \mathbf{g}_i^{t,k}, \quad \mathbf{w}_i^{t,0} = \mathbf{w}^{t-1}, \quad k = 0, \dots, K-1. \quad (\text{E.1})$$

Here $\mathbf{g}_i^{t,k}$ is an unbiased estimator of $\nabla F_i(\mathbf{w}_i^{t,k})$ (Assumption 2). After K steps, the server aggregates

$$\mathbf{w}^t = \sum_{i=1}^m \alpha_i^t \mathbf{w}_i^{t,K}. \quad (\text{E.2})$$

Step 1: A one-round descent inequality for F^t . Since each F_i is L -smooth (Assumption 1), the weighted objective $F^t(\cdot) = \sum_i \alpha_i^t F_i(\cdot)$ is also L -smooth. Applying (A.1) to F^t at $(\mathbf{w}^{t-1}, \mathbf{w}^t)$ gives

$$F^t(\mathbf{w}^t) \leq F^t(\mathbf{w}^{t-1}) + \langle \nabla F^t(\mathbf{w}^{t-1}), \mathbf{w}^t - \mathbf{w}^{t-1} \rangle + \frac{L}{2} \|\mathbf{w}^t - \mathbf{w}^{t-1}\|^2. \quad (\text{E.3})$$

Define $\Delta^t := \mathbf{w}^t - \mathbf{w}^{t-1}$. From (E.1)–(E.2),

$$\mathbf{w}_i^{t,K} - \mathbf{w}^{t-1} = -\eta \sum_{k=0}^{K-1} \mathbf{g}_i^{t,k}, \quad \Delta^t = -\eta \sum_{k=0}^{K-1} \sum_{i=1}^m \alpha_i^t \mathbf{g}_i^{t,k}.$$

Plugging into (E.3) yields

$$F^t(\mathbf{w}^t) \leq F^t(\mathbf{w}^{t-1}) - \eta \sum_{k=0}^{K-1} \left\langle \nabla F^t(\mathbf{w}^{t-1}), \sum_{i=1}^m \alpha_i^t \mathbf{g}_i^{t,k} \right\rangle + \frac{L}{2} \|\Delta^t\|^2. \quad (\text{E.4})$$

Take expectation over the stochasticity (conditioning on the history up to $\mathbf{w}^{t-1}$). Using unbiasedness,

$$\mathbb{E}[\mathbf{g}_i^{t,k} | \mathbf{w}_i^{t,k}] = \nabla F_i(\mathbf{w}_i^{t,k}), \quad \mathbb{E} \left[\sum_{i=1}^m \alpha_i^t \mathbf{g}_i^{t,k} \right] = \mathbb{E} \left[\sum_{i=1}^m \alpha_i^t \nabla F_i(\mathbf{w}_i^{t,k}) \right].$$

Add and subtract $\sum_i \alpha_i^t \nabla F_i(\mathbf{w}^{t-1}) = \nabla F^t(\mathbf{w}^{t-1})$:

$$\begin{aligned} \left\langle \nabla F^t(\mathbf{w}^{t-1}), \sum_i \alpha_i^t \nabla F_i(\mathbf{w}_i^{t,k}) \right\rangle &= \|\nabla F^t(\mathbf{w}^{t-1})\|^2 \\ &+ \left\langle \nabla F^t(\mathbf{w}^{t-1}), \sum_i \alpha_i^t (\nabla F_i(\mathbf{w}_i^{t,k}) - \nabla F_i(\mathbf{w}^{t-1})) \right\rangle. \end{aligned} \quad (\text{E.5})$$

Using Cauchy–Schwarz and Young’s inequality (A.2) with $\rho = 1$,

$$\begin{aligned} &\left\langle \nabla F^t(\mathbf{w}^{t-1}), \sum_i \alpha_i^t (\nabla F_i(\mathbf{w}_i^{t,k}) - \nabla F_i(\mathbf{w}^{t-1})) \right\rangle \\ &\geq -\frac{1}{2} \|\nabla F^t(\mathbf{w}^{t-1})\|^2 - \frac{1}{2} \left\| \sum_i \alpha_i^t (\nabla F_i(\mathbf{w}_i^{t,k}) - \nabla F_i(\mathbf{w}^{t-1})) \right\|^2. \end{aligned} \quad (\text{E.6})$$

By convexity of squared norm (A.3) and L -smoothness,

$$\begin{aligned} \left\| \sum_i \alpha_i^t (\nabla F_i(\mathbf{w}_i^{t,k}) - \nabla F_i(\mathbf{w}^{t-1})) \right\|^2 &\leq \sum_i \alpha_i^t \|\nabla F_i(\mathbf{w}_i^{t,k}) - \nabla F_i(\mathbf{w}^{t-1})\|^2 \\ &\leq L^2 \sum_i \alpha_i^t \|\mathbf{w}_i^{t,k} - \mathbf{w}^{t-1}\|^2. \end{aligned} \quad (\text{E.7})$$

Combining (E.5)–(E.7) inside (E.4), summing over $k = 0, \dots, K-1$, and taking expectation gives

$$\begin{aligned} \mathbb{E}[F^t(\mathbf{w}^t)] &\leq \mathbb{E}[F^t(\mathbf{w}^{t-1})] - \frac{\eta}{2} \sum_{k=0}^{K-1} \mathbb{E} \left[\|\nabla F^t(\mathbf{w}^{t-1})\|^2 \right] \\ &+ \frac{\eta L^2}{2} \sum_{k=0}^{K-1} \mathbb{E} \left[\sum_i \alpha_i^t \|\mathbf{w}_i^{t,k} - \mathbf{w}^{t-1}\|^2 \right] + \frac{L}{2} \mathbb{E}[\|\Delta^t\|^2]. \end{aligned} \quad (\text{E.8})$$

Step 2: Bounding local drift and the aggregated update. From (E.1),

$$\mathbf{w}_i^{t,k} - \mathbf{w}^{t-1} = -\eta \sum_{s=0}^{k-1} \mathbf{g}_i^{t,s}.$$

By Jensen and (A.4) with $\mathbf{a} = \nabla F_i(\mathbf{w}^{t-1})$,

$$\begin{aligned} \mathbb{E} \left[\left\| \mathbf{w}_i^{t,k} - \mathbf{w}^{t-1} \right\|^2 \right] &\leq \eta^2 k \sum_{s=0}^{k-1} \mathbb{E} \left[\left\| \mathbf{g}_i^{t,s} \right\|^2 \right] \\ &\leq 2\eta^2 k \sum_{s=0}^{k-1} \mathbb{E} \left[\left\| \mathbf{g}_i^{t,s} - \nabla F_i(\mathbf{w}_i^{t,s}) \right\|^2 \right] + 2\eta^2 k \sum_{s=0}^{k-1} \mathbb{E} \left[\left\| \nabla F_i(\mathbf{w}_i^{t,s}) \right\|^2 \right]. \end{aligned} \quad (\text{E.9})$$

Assumption 2 gives $\mathbb{E} \left[\left\| \mathbf{g}_i^{t,s} - \nabla F_i(\mathbf{w}_i^{t,s}) \right\|^2 \right] \leq \sigma^2$. Also, by smoothness and (A.4),

$$\left\| \nabla F_i(\mathbf{w}_i^{t,s}) \right\|^2 \leq 2 \left\| \nabla F_i(\mathbf{w}^{t-1}) \right\|^2 + 2L^2 \left\| \mathbf{w}_i^{t,s} - \mathbf{w}^{t-1} \right\|^2.$$

Using $\eta \leq \frac{1}{4LK}$ to close the recursion (standard in local-SGD analyses), we obtain

$$\sum_{k=0}^{K-1} \sum_{i=1}^m \mathbb{E} \left[\left\| \mathbf{w}_i^{t,k} - \mathbf{w}^{t-1} \right\|^2 \right] \leq 4\eta^2 K^2 \mathbb{E} \left[\left\| \nabla F^t(\mathbf{w}^{t-1}) \right\|^2 \right] + 4\eta^2 K^2 \zeta^2 + 4\eta^2 K^2 \sigma^2. \quad (\text{E.10})$$

Next, for $\Delta^t = -\eta \sum_{k=0}^{K-1} \sum_{i=1}^m \alpha_i^t \mathbf{g}_i^{t,k}$, a similar bound yields

$$\mathbb{E} \left[\left\| \Delta^t \right\|^2 \right] \leq 4\eta^2 K^2 \mathbb{E} \left[\left\| \nabla F^t(\mathbf{w}^{t-1}) \right\|^2 \right] + 4\eta^2 K^2 \zeta^2 + 4\eta^2 K^2 \sigma^2. \quad (\text{E.11})$$

Step 3: finish the one-round descent. Plugging (E.10) and (E.11) into (E.8) and using $\eta \leq \frac{1}{4LK}$ to keep the gradient term on the left gives

$$\mathbb{E}[F^t(\mathbf{w}^t)] \leq \mathbb{E}[F^t(\mathbf{w}^{t-1})] - \frac{\eta K}{4} \mathbb{E} \left[\left\| \nabla F^t(\mathbf{w}^{t-1}) \right\|^2 \right] + 2L\eta K \sigma^2 + 2L^2 \eta^2 K^2 \zeta^2. \quad (\text{E.12})$$

Rearranging yields

$$\mathbb{E} \left[\left\| \nabla F^t(\mathbf{w}^{t-1}) \right\|^2 \right] \leq \frac{4}{\eta K} \left(\mathbb{E}[F^t(\mathbf{w}^{t-1})] - \mathbb{E}[F^t(\mathbf{w}^t)] \right) + 8L\eta K \sigma^2 + 8L^2 \eta^2 K^2 \zeta^2. \quad (\text{E.13})$$

Step 4: Telescope across rounds with objective drift. Summing (E.13) over $t = 1, \dots, R$ and dividing by R ,

$$\begin{aligned} \frac{1}{R} \sum_{t=1}^R \mathbb{E} \left[\left\| \nabla F^t(\mathbf{w}^{t-1}) \right\|^2 \right] &\leq \frac{4}{\eta K R} \sum_{t=1}^R \left(\mathbb{E}[F^t(\mathbf{w}^{t-1})] - \mathbb{E}[F^t(\mathbf{w}^t)] \right) \\ &\quad + 8L\eta K \sigma^2 + 8L^2 \eta^2 K^2 \zeta^2. \end{aligned} \quad (\text{E.14})$$

To handle the non-telescoping part due to changing F^t , write

$$\begin{aligned} \sum_{t=1}^R (F^t(\mathbf{w}^{t-1}) - F^t(\mathbf{w}^t)) &= F^1(\mathbf{w}^0) - F^R(\mathbf{w}^R) \\ &\quad + \sum_{t=2}^R (F^t(\mathbf{w}^{t-1}) - F^{t-1}(\mathbf{w}^{t-1})). \end{aligned} \quad (\text{E.15})$$

Taking expectation, applying Lemma 3, and using $F^R(\mathbf{w}^R) \geq F_{\inf}$, we obtain

$$\sum_{t=1}^R \left(\mathbb{E}[F^t(\mathbf{w}^{t-1})] - \mathbb{E}[F^t(\mathbf{w}^t)] \right) \leq (F^1(\mathbf{w}^0) - F_{\inf}) + B_F \sum_{t=2}^R \mathbb{E} \left[\left\| \alpha^t - \alpha^{t-1} \right\|_1 \right]. \quad (\text{E.16})$$

Plugging (E.16) into (E.14) gives the desired result:

$$\begin{aligned} \frac{1}{R} \sum_{t=1}^R \mathbb{E} \left[\left\| \nabla F^t(\mathbf{w}^{t-1}) \right\|^2 \right] &\leq \frac{4(F^1(\mathbf{w}^0) - F_{\inf})}{\eta K R} + 8L\eta K \sigma^2 + 8L^2 \eta^2 K^2 \zeta^2 \\ &\quad + \frac{4B_F}{\eta K} \cdot \frac{1}{R} \sum_{t=2}^R \mathbb{E} \left[\left\| \alpha^t - \alpha^{t-1} \right\|_1 \right]. \end{aligned} \quad (\text{E.17})$$

□

Data availability

Data will be made available on request.

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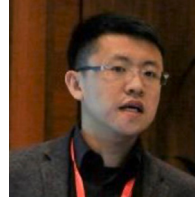
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